

Problems on Ceramic Materials

1.- For the following Zinc Blende structure (ZnS) calculate:

- The volume of the unit cell and volumetric density expressed in g/cm^3 .
- The packing factor.
- The linear density (ions/mm) in the direction $[111]$.
- The planar density (ions/ mm^2) in the plane (111) .

Data: $r(\text{Zn}^{2+}) = 0.060 \text{ nm}$; $r(\text{S}^{2-}) = 0.174 \text{ nm}$, $M(\text{Zn}) = 63.54$, $M(\text{S}) = 32.06$.

(Solution: a) $V = 1.57 \times 10^{-22} \text{ cm}^3$ $\rho_v = 4.04 \text{ g/cm}^3$; b) $f = 0.585$; c) $\rho[111] = 2.13 \times 10^6 \text{ ions/mm}$; d) $\rho(111) = 7.92 \times 10^{12} \text{ ions/mm}^2$)

2- The Perovskite structure (CaTiO_3) can be described as an FCC packing of Ca^{2+} and O^{2-} ions on which Ti^{4+} occupies $\frac{1}{4}$ of the octahedral sites. Calculate:

- The volume of the unit cell.
- The packing factor.
- The linear density (ions/nm) in the direction $[111]$.
- The planar density (ions/ nm^2) in the plane (111) .
- The volumetric density expressed in g/cm^3 .

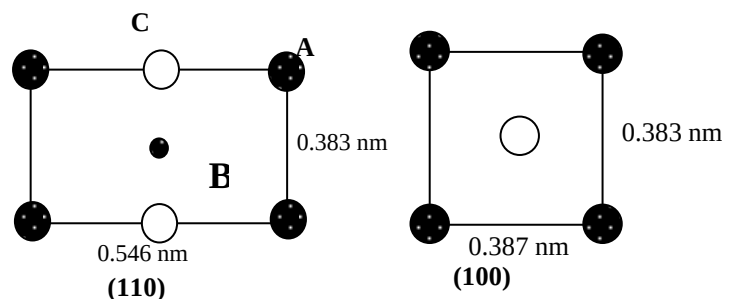
Data: $r(\text{Ca}^{2+}) = 0.106 \text{ nm}$; $r(\text{O}^{2-}) = 0.132 \text{ nm}$; $r(\text{Ti}^{4+}) = 0.036 \text{ nm}$; $M(\text{Ca}) = 40.08$; $M(\text{O}) = 16.00$; $M(\text{Ti}) = 47.90$.

(Solution: a) $V = 0.038 \text{ nm}^3$; b) $f = 0.897$; c) $\rho[111] = 3.44 \text{ ions/nm}$; d) $\rho(111) = 20.4 \text{ ions/nm}^2$; e) $\rho_v = 5.95 \text{ g/cm}^3$)

3.- In the figure, two crystallographic planes are represented. They correspond to a unit cell of a ceramic material with the following formula

$\text{A}_x\text{B}_y\text{C}_z$. Determine:

- The crystalline system they belong to
- The stoichiometry and density (g/cm^3)
- The structure type and the description of the structure in terms of the closed packing sequence.



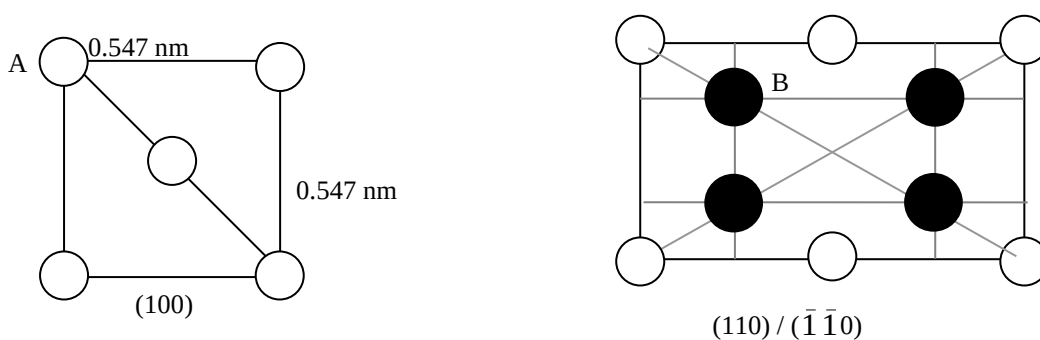
Data: $M(\text{A}) = 40 \text{ g/mol}$, $M(\text{B}) = 48 \text{ g/mol}$, $M(\text{C}) = 16 \text{ g/mol}$.

(Solution: a) orthorhombic symmetry; b) ABC_3 ; $\rho = 3.96 \text{ g/cm}^3$, c) Perovskite-type structure)

4.- In the figure, two crystallographic planes are represented. They correspond to a cubic unit cell of a ceramic material with the following formula A_xB_y . Determine.

- The structure type and its description (**Solution: Fluorite type structure, CaF_2 . FCC close packing of A ions. The B atoms occupy all the tetrahedral sites**)
- The stoichiometry of the compound (**Solution: AB_2**)
- The density expressed in g/cm^3 (**Solution: $10.95 g/cm^3$**)
- The packing factor (**Solution: $f = 0.589$**)

Data: $M(A) = 238 g/mol$, $M(B) = 16 g/mol$, $r(A) = 0.105 nm$, $r(B) = 0.132 nm$.

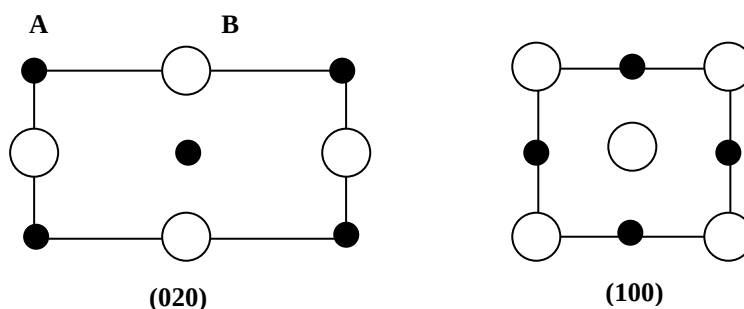


5.- In the figure, two crystallographic planes are represented. They correspond to a cubic unit cell of a ceramic material with the following formula AB. If the diagonal of the cube is 0.824 nm, determine.

- The structure type and its description.
- The stoichiometry of the compound.
- The density expressed in g/cm^3 .
- The packing factor.

Data: $M(A) = 40 g/mol$, $M(B) = 16 g/mol$, $r(A) = 0.106 nm$.

(Solution: a) NaCl type; b) AB; c) $3.45 g/cm^3$; d) 0.54).



6.-The compound SrSnO_3 has a perovskite structure, with Sn^{4+} occupying the octahedral interstitial position at the center of the cube. DATA: Atomic masses: O = 16, Sr = 87.6, Sn = 118.7; Ionic radii: $\text{O}^{2-} = 0,146 \text{ nm}$, $\text{Sn}^{4+} = 0,071 \text{ nm}$, $\text{Sr}^{2+} = 0,113 \text{ nm}$. Calculate:

- a) The linear density in ions/nm along the $[101]$ direction
- b) The surface density in ions/ nm^2 in the (101) plane
- c) The mass of a cylindrical rod with a 5 mm radius and a 10 mm length

(Solution: a) $\rho_{[101]} = 3.86 \text{ ions/nm}$; b) $\rho_{(101)} = 15.87 \text{ ions/nm}^2$; c) $m = 6.76 \text{ g}$)

7- Considering that MgO crystallizes in a NaCl-type structure, determine:

DATA: $r(\text{Mg}^{2+}) = 0,078 \text{ nm}$; $r(\text{O}^{2-}) = 0,132 \text{ nm}$; $M(\text{MgO}) = 40 \text{ g/mol}$; $N_A = 6,023 \cdot 10^{23} \text{ átomos/mol}$

- a) The volume of the unit cell
- b) The packing factor
- c) The linear density (in ions/nm) in the $[111]$ direction
- d) The planar density (in ions/ nm^2) in the (111) plane
- e) The volumetric density (in g/cm^3)

(Solution: a) $V = 0.074 \text{ nm}^3$; b) $f = 0.63$; c) $\rho_{[111]} = 2.75 \text{ ions/nm}$; d) $\rho_{(111)} = 13.1 \text{ ions/nm}^2$; e) $\rho_v = 3.59 \text{ g/cm}^3$)